

Characterization of direct Purkinje cell outputs to the brainstem

Christopher H Chen , Zhiyi Yao, Shuting Wu, Wade G Regehr 

Department of Neurobiology, Harvard Medical School, Boston, United States • Department of Neural and Behavioral Sciences, The Pennsylvania State University, Hershey, United States

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This **important** paper provides an unbiased landscape for the cerebellar cortical outputs to the brainstem nuclei. By conducting anatomical and physiological analyses of the axonal terminals of Purkinje cells, the data provides **convincing** evidence that Purkinje cells innervate brainstem nuclei directly. The results show that in addition to previously known inputs to vestibular and parabrachial nuclei, Purkinje cells synapse onto the pontine central grey nucleus but have little effect on the locus coeruleus and mesencephalic trigeminal neurons.

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Abstract

Purkinje cells (PCs) primarily project to cerebellar nuclei but also directly innervate the brainstem. Some PC-brainstem projections have been described previously, but most have not been thoroughly characterized. Here we use a PC-specific cre line to anatomically and electrophysiologically characterize PC projections to the brainstem. PC synapses are surprisingly widespread, with the highest densities found in the vestibular and parabrachial nuclei. However, there are pronounced regional differences in synaptic densities within both the vestibular and parabrachial nuclei. Large optogenetically-evoked PC-IPSCs are preferentially observed in subregions with the highest densities of PC synapses, suggesting that PCs selectively influence these areas and the behaviors they regulate. Unexpectedly, the pontine central gray and nearby subnuclei also contained a low density of PC synapses, and large PC-IPSCs are observed in a small fraction of cells. We combined electrophysiological recordings with immunohistochemistry to assess the molecular identities of two putative PC targets: PC synapses onto mesencephalic trigeminal neurons were not observed even though these cells are in close proximity to PC boutons. PC synapses onto locus coeruleus neurons are exceedingly rare or absent, even though previous studies concluded that PCs are a major input to these neurons. The availability of a highly selective cre line for PCs allowed us to study functional synapses, while avoiding complications that can accompany the use of viral approaches. We conclude that PCs directly innervate numerous brainstem nuclei, but only inhibit a small fraction of cells in many nuclei. This suggests that PCs target cell types with specific behavioral roles in brainstem regions.

Introduction

Delineating the PC output pathways that mediate signals from the cerebellar cortex to the rest of the brain is a vital step in understanding cerebellar function. Although most studies of PC outputs have focused on the prominent connections to cerebellar nuclei ^{1,2}, PCs also project directly to the brainstem. Extensive PC projections to vestibular nuclei ³⁻⁶ play vital roles in vestibular and ocular-related functions ⁷⁻⁹. PCs also project to the prepositus hypoglossal nucleus, and this pathway may play a role in gaze stabilization ¹⁰⁻¹³. In addition, PCs prominently project to the parabrachial nucleus, which is suited to regulate broad-ranging nonmotor and autonomic functions ¹⁴⁻¹⁸.

Several studies suggested that PCs also inhibit cells in other brainstem nuclei that could allow the cerebellum to control diverse behaviors. Rabies-based tracing approaches suggested that PCs are the primary inhibitory inputs to the locus coeruleus (LC) ¹⁹⁻²¹, but this is controversial. These studies have been highly influential and raised the possibility that PCs directly modulate LC-norepinephrine release, and regulate stress, attention, motivation, arousal, pain modulation, reward processing and other behaviors ²²⁻²⁷. Furthermore, it was also suggested that disruption of the PC-LC pathway contributes to ASD and ADHD ²⁸, and impairs the ability of the cerebellum to control seizures ²⁹. However, other studies do not see strong evidence for this PC-LC connection. Retrograde tracing studies of projections to the LC using dye tracers observed minimal PC labelling ³⁰⁻³², and PC axons largely avoid the LC ¹⁷. Thus, there is no consensus on whether PCs strongly inhibit LC neurons. There is also suggestive evidence that the cerebellum influences additional brainstem nuclei: the injection of trans-synaptic viruses into the posterior vermis labelled cells in pontine central grey (PCG) and nearby nuclei including the LC, Barrington's nucleus, dorsal tegmentum, and others ¹⁷. These findings raise the possibility that direct PC projections could allow the cerebellum to regulate arousal, sleep-state transitions, micturition, and other behaviors ³³⁻³⁷. However, the viruses used in these studies can also retrogradely label cells ^{38,39}, and it is therefore necessary to assess whether there are functional PC synapses in these nuclei.

In this study, we determine whether PCs make synapses and inhibit cells within various brainstem nuclei by determining whether PC synaptic boutons are present, and if PCs make functional synapses. Using *PCP2-cre/Ai34D (PC/synaptophysin-tdTomato)* mice to label PC synapses and map their locations, we find that PCs project widely in the brainstem, and that synaptic densities are highly nonuniform within different nuclei and subnuclei. We identify and characterize PC synapses in brainstem nuclei using optogenetics and *PCP2-cre/Ai32 (PC/ChR2-YFP)* mice. There was good agreement between the density of PC synaptic boutons and the probability of evoking PC-IPSCs in neurons within different regions and subregions. In regions with a low density of PC boutons, PC-IPSCs were typically observed in a small fraction of cells, raising the possibility that PCs selectively target specific types of neurons in these regions. These findings provide an important resource that will allow future studies to clarify the PC targets in various brainstem nuclei and the behaviors controlled by these output pathways.

Results

Visualization of PC projections to the brainstem

We visualized PC synapses in the brainstem of *PC/synaptophysin-tdTomato* mice in which synaptic boutons are intensely labelled, and axons, dendrites and somata weakly labelled. We imaged synaptophysin-tdTomato in both sagittal (**Fig. 1A**) and coronal sections (**Fig. 1B**), and aligned

the images to the Allen Common Coordinate Framework (CCF) [40](#) based on the contours of the tissue. We collapsed the vestibular subnuclei (MV, LAV etc.) into the VN, and labelled the territories encompassing the dorsomedial pons as simply the PCG (see Methods). Likewise, the rostral/caudal or medial/lateral divisions of other nuclei are not annotated specifically (PGRN, NTS, etc.).

In low magnification images (**Fig. 1A, 1B**, [left images](#)), labeling is observed in the cerebellar nuclei and brainstem. In sagittal views, brainstem labeling is sparser in the medial sections than in lateral sections. In coronal sections, the brainstem labeling is densest at bregma -5.85 mm, and sparser in anterior and posterior sections. Medium (**Fig. 1A, 1B**, [2nd column from left](#)) and high magnification images (**Fig. 1A, 1B i-iv) showed highly variable densities of PC boutons in different regions. In some regions, labelling was extremely faint for medium magnification images, but PC synapses were apparent at higher magnifications (for example, in the PCG in **Fig. 1Aa i** and **Fig. 1Ba ii**).**

There were pronounced regional differences in the extent of labelling within the VN and the PB. In sagittal sections, for the VN the densest PC labelling was observed ventral to the cerebellar nuclei and the fourth ventricle (**Fig. 1Ac ii**), and expression fanned out ventrally, anteriorly, and posteriorly. Dense labeling of the VN was also present in the anterior dorsal regions of the other sagittal slices (**Fig. 1Aa ii**, **Fig. 1Ab ii+iii**, **Fig. 1Ac ii**, **Fig. 1Ad ii**). Although PC boutons were present throughout the VN, in anterior regions the densities were low near the midline (**Fig. 1Aa iii**, **Fig. 1Ab iv**) and moderate densities in more lateral sections (**Fig. 1Ac iv**, **Fig. 1Ad iv**). The density gradient of PC synapses within the VN was more apparent in coronal sections. Synaptic densities were highest in dorsal VN (**Fig. 1Bb i+ii**, **Fig. 1Bc i**, **Fig. 1Bd i**, **Fig. 1Be ii**), and lower in ventral VN (**Fig. 1Ba iii**, **Fig. 1Bb iii**, **Fig. 1Bc iii**, **Fig. 1Bd ii**). A gradient of labelling was also apparent within the PB. The highest densities were present in posterior PB (**Fig. 1Ac i**, **Fig. 1Ba i**). The differences in the densities of PC synapses within subregions of the VN and the PB suggest that PCs preferentially regulate specific subregions.

Low levels of labelling were apparent in many other brainstem regions. In the PCG, fluorescence is apparent in high magnification views (**Fig. 1Aa i**, **Fig. 1Ba ii+iv**). Sparse labelling is present in the PRP (**Fig. 1Ba-d iv**), which is consistent with previous studies [10-13](#). Very little labelling was found in the GRN (**Fig. 1Aa iv**), the PRNr (**Fig. 1Ad iii**), the LC (**Fig. 1Ab i**), and the NTS (**Fig. 1Be iii**).

These fluorescence images establish that there is considerable variation in the densities of PC synapses between and within various brainstem regions. The highest densities of PC synapses are present in the range bregma -5.85 mm to -5.95 mm, and 1.5 to 2 mm from the midline. Notably PC synapses were only observed in the dorsal region of the brainstem.

Quantification of PC synapses in the brainstem

To quantify the densities of PC synapses in the brainstem, we detected individual PC boutons, and registered images to the Allen CCF. Experiments were performed in *CGRP-GFP/PCP2-cre/synaptophysin-tdTomato* mice. We used synaptophysin-tdTomato fluorescence to label PC synapses, vGAT immunohistochemistry to visualize all GABAergic synapses, and CGRP-GFP to label subnuclei to aid in atlas alignment. The resulting alignment is shown for a coronal slice (**Fig. 2A**, see Methods). To quantify synapses, confocal images were taken for a z-stack of 1.5 μ m of the dorsal brainstem bilaterally. All regions previously observed with tdTomato signal (**Fig 1**) were imaged, as well as additional sections with no obvious labeling within 100 μ m of labelled regions. Since weak tdTomato labeling is observed in the axons of PCs, vGAT immunohistochemistry was used to help identify putative pre-synaptic sites. In addition, PC contributions to total inhibition in each region can be measured by comparing GABAergic boutons (**Fig. 2B**, [lower right, grey dots](#)) with or without tdTomato colabelling (**Fig. 2B**, [lower right, red dots](#), see Methods). This approach identifies putative presynaptic sites but does not identify functional synapses.

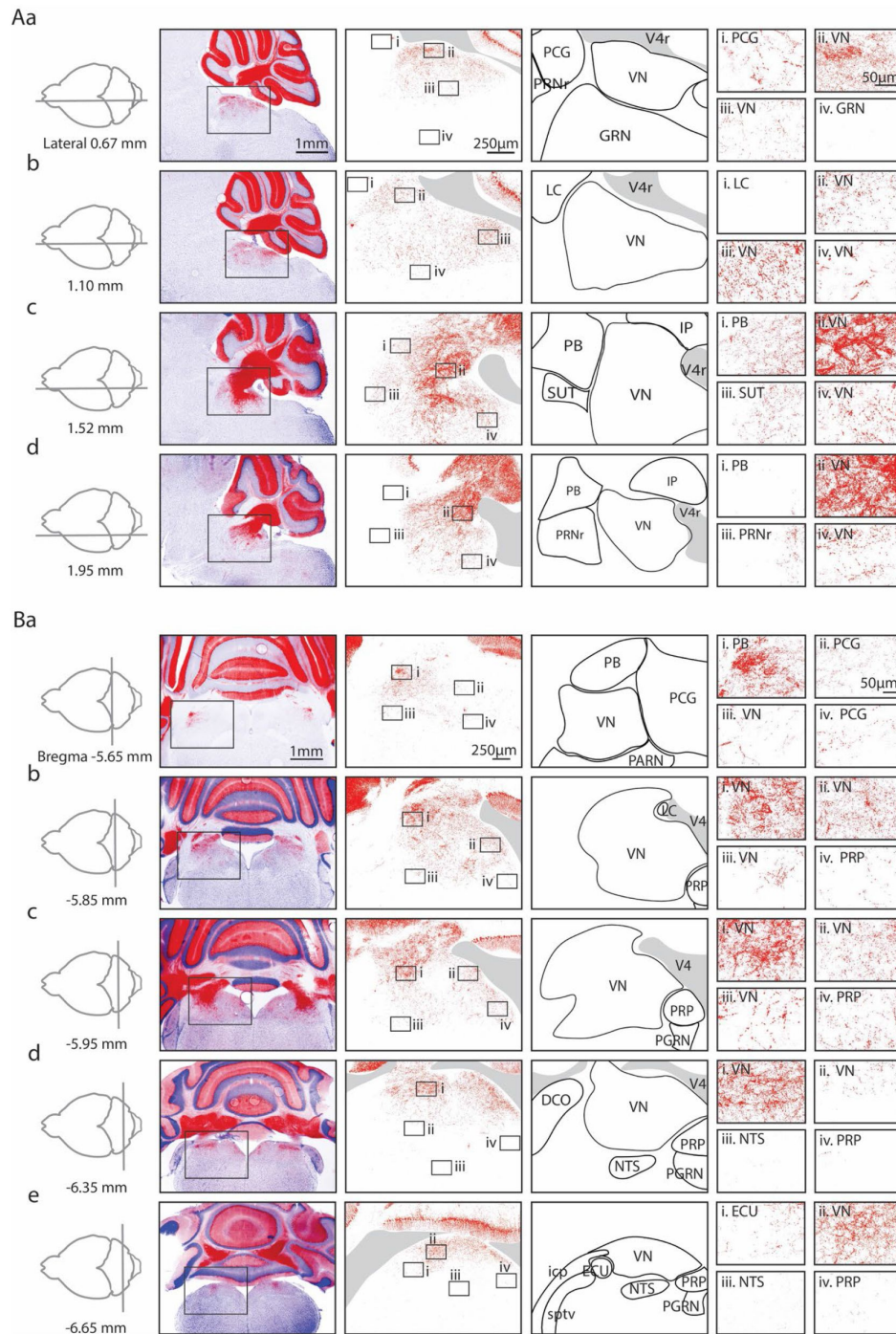


Figure 1

Anatomical characterization of Purkinje cell outputs to the brainstem using *PC/synaptophysin-tdTomato* mice.

Aa. First column. Diagram showing parasagittal slice location.

Second column. Low magnification views of tdTomato (tdT) fluorescence (red) in the cerebellar cortex and brainstem.

Third column. Medium magnification views of the regions indicated in Ab.

Fourth column. Anatomical regions corresponding to the preceding column.

Last column. i-iv. High magnification views of the regions indicated in the preceding column. Ab-d. As in Aa, but for subsequent parasagittal slices.

B. As in A, but for coronal slices.

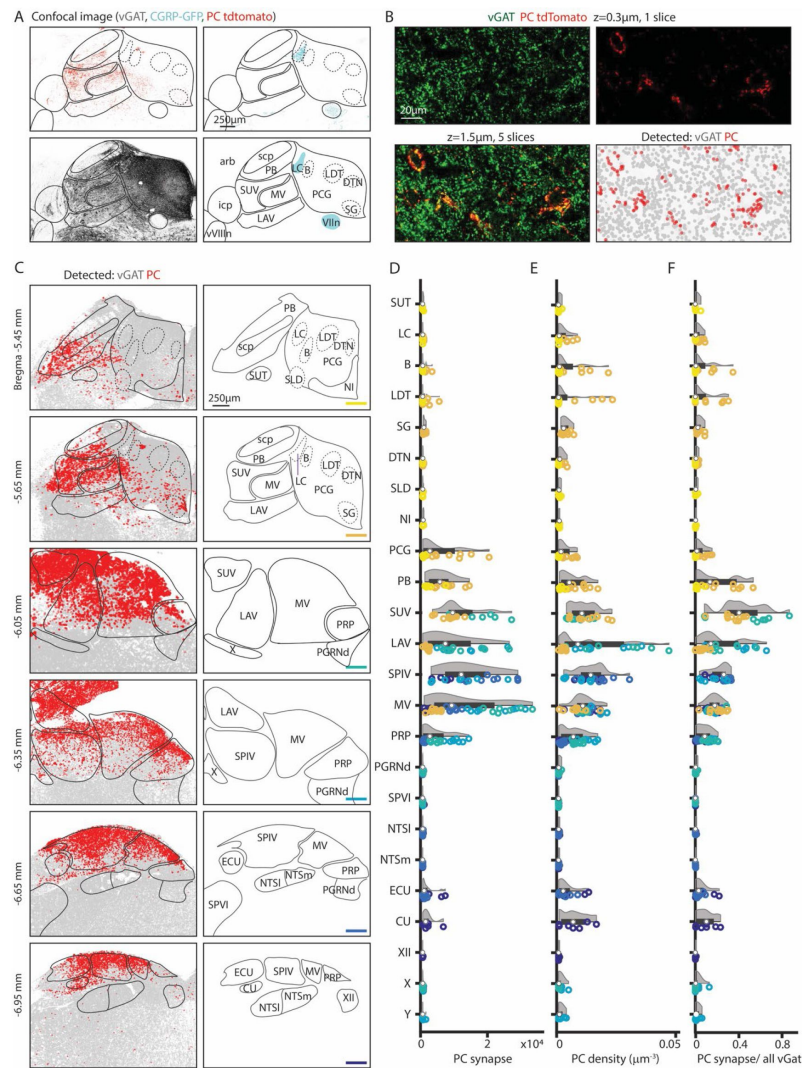


Figure 2

Quantification of Purkinje cell synapses in the brainstem.

GABAergic boutons were detected using a vGAT antibody and PC presynaptic boutons were detected based on tdTomato labelling in *PC/synaptophysin-tdTomato* mice.

A. (upper left) Confocal image of td-Tomato fluorescence.

(upper right) GFP fluorescence in CGRP-GFP mouse in which the locus coeruleus and facial nerve are labelled. (lower left) vGAT labelling of GABAergic boutons.

(lower right) Allen Atlas coronal section corresponding to confocal images with landmarks used for registration highlighted in cyan.

B. (upper left) vGAT labelling in a single confocal section. (upper right) tdT labelling in a single confocal section.

(lower left) z-stack of vGAT and tdT labelling in 5 sections (1.5 μm thick).

(lower right) Detected GABAergic boutons based on vGAT labelling (grey) and PC boutons (red).

C. (left) Detected GABAergic boutons corresponding to PCs (red), and neurons other than PCs (grey) from six example coronal sections.

(right) Labelled anatomic regions for each coronal section.

D. Number of PC synapses detected per section in each region. Individual section values are represented with symbol colors correspond to the sections in C, right (bottom right corner color bars). Violin plot of the region shown in grey with average and quartile values in black.

E. Density of PC boutons in each region. Individual section and region average as represented in D.

F. Percentages of GABAergic boutons that correspond to PCs for each region. Individual section and region average as represented in D.

This approach was used to detect putative PC boutons in a series of coronal slices from Bregma -5.45 to -6.95, where the majority of PC projections reside. Inhibitory synapses were present at high densities in all regions in all slices (**Fig. 2C**, left, grey), while PC synapses were spatially restricted (**Fig. 2C**, left, red). The total number of detected PC boutons, the density of PC synapses, and the fraction of inhibitory boutons that are made by PCs, were determined for each region (**Fig. 2D-F**, data points color-coded to indicate slice position, $n = 3$ mice, 2 hemispheres/mouse). Areas with observable PC boutons were selected for display along with their near neighbors, including example areas with no tdTomato labeling (PGRNd, SPVI, NTSL, NTSm). Areas ventral to displayed regions were not labelled.

High densities of PC synapses are present in the superior, lateral, spinal and medial vestibular nuclei, but there are pronounced regional differences in bouton densities within the subnuclei and related nuclei (**Fig. 2C-F**). In the superior vestibular nucleus, PC synapses are present at high density and comprise up to 85% of all inhibitory inputs. In the lateral vestibular nucleus, the number, density, and fractional contribution of PC synapses, were largest in posterior sections. PC inputs contributed to ~40% of all inhibitory inputs in posterior LAV. The medial vestibular nucleus spanned the longest anterior-posteriorly and has the most PC synapses in the middle sections, from bregma -6.05 mm to -6.35 mm. There was a pronounced dorsal-ventral gradient in the density of PC synapses. The spinal vestibular nucleus is the most posterior nucleus and had less variability across the anterior-posterior axis. It has a high average PC synapse number and density across sections and 26% of inhibition originates from PCs. The prepositus nucleus also receives PC inputs that are highest at bregma -6.05 mm and -6.35 mm and highest in dorsal regions. PCs contribute up to 9% of the total inhibitory inputs in the prepositus nucleus (**Fig. 2D** left, middle, right). In the more posterior sections (bregma -6.65 mm and -6.95 mm), PC synapses were restricted to the dorsal region in the vestibular nuclei, the external cuneate, and the cuneate. The external cuneate and the cuneate are small regions, and the densities were correspondingly higher though the absolute number of synapses are relatively low and variable. The external cuneate spanned from bregma -6.65mm to -6.95 but has more PC synapses in the posterior section. It is evident that the vestibular nuclei, the cuneate nucleus and prepositus nucleus are not uniformly targeted by PCs.

There is a posterior to anterior gradient of PC synapses in the PB. In anterior sections (**Fig. 2C**, bregma -5.45 mm to -5.65 mm) high densities of PC synapses were only apparent in a few localized regions of the PB. The medial-ventral region of the PB (bregma -5.65 mm) has extensive PC inputs, comparable in density to some regions of the vestibular nuclei.

PC synapses are present but at low densities within the pontine central gray (PCG) and nearby subnuclei including the LC, Barrington's nucleus, sublateral dorsal nucleus, lateral dorsal tegmental nucleus, dorsal tegmental nucleus, and supragenual nucleus (**Fig. 2C**, bregma -5.45 mm to -5.65 mm). PC synapses make up a small fraction of inhibitory synapses in these areas (**Fig. 2F**). The Barrington's nucleus and the laterodorsal tegmental nucleus had a higher PC density and fraction of inhibition, although it is important to note that these PCG subregions are relatively small in area. The LC, which is labelled in CGRP mice, had few PC synapses with a slightly elevated PC density count in a few posterior sections.

Characterization of functional properties of PC synapses in the brainstem

To confirm these PC connections in different brainstem regions and examine their strength, we optically stimulated PC fibers in *PC/ChR2-YFP* mice and measured IPSCs in coronal brain slices of the dorsal brainstem. It is likely that this approach provides a good estimate of connectivity even when PC axons are cut, because severed axons expressing ChR2 can be optogenetically activated to evoke synaptic responses^{41,44}. We recorded optically-evoked IPSCs in 133 of 310 neurons in the dorsal brainstem (between bregma -5.45 mm to -6.35 mm) to generate a physiological map of

PC inputs to the brainstem. The IPSC amplitudes of the recorded brainstem neurons (**Fig. 3A** [↗](#), greyscale color-coded) are shown along with the density maps of PC boutons from **Fig. 2** [↗](#). For the density map, PC bouton counts were binned into $144 \times 144 \times 4 \mu\text{m}$ volumes, and averaged across the multiple sections. The likelihood of optically-evoking a PC-IPSC varied with coronal section from anterior to posterior was 9%, 32%, 51%, and 45% (-5.45, -5.65, -6.05 and -6.35 mm to Bregma). We quantified the response amplitudes (**Fig. 3B** [↗](#)), probability of evoking a PC-IPSC (**Fig. 3C** [↗](#)), and the IPSC amplitudes (**Fig. 3D** [↗](#)).

In the most anterior slices (Bregma -5.45 mm and -5.65 mm), PC-IPSCs were not evoked at most locations (47/53 cells nonresponding), which is consistent with the very low density of PC synaptic boutons (**Fig. 3A** [↗](#)). For the PCG, no PC-IPSCs were evoked in the Bregma -5.45 mm section. In the Bregma -5.65 mm section PC-IPSCs were evoked in 23% of cells, with highly variable amplitudes (**Fig. 3B** [↗](#)), and the average PC-IPSC in responding cells was over 1 nA (**Fig. 3D** [↗](#)). For the PB, PC-IPSCs were evoked with a low response probability in Bregma -5.45 mm slices, but at Bregma -5.65 mm, 48% of cells responded (**Fig. 3C** [↗](#)). In the PB, the amplitudes of IPSCs onto different cells were highly variable (50 pA to over 10 nA) (**Fig. 3B** [↗](#)), and the average of responding cells was 2 nA.

In more posterior slices (Bregma -6.05 mm and -6.35), PC-IPSCs were evoked in a high percentage of cells in the SUV, LAV, MV, and PRP (45%, 59%, 58%, and 52% respectively, **Fig. 3C** [↗](#)). The SUV and LAV had predominantly large IPSCs in responding cells, with most evoked currents larger than 500 pA and the average PC-IPSC of responding cells was several nanoamps. In the SPIV, MV, and PRP, PC-IPSCs ranged from 50 pA to ~2nA and average IPSC amplitudes of responding cells was less than 1 nA (**Fig. 3B, D** [↗](#)). Overall, PC-IPSCs were evoked in a larger percentage of cells in regions with a high density of PC boutons (**Fig. 3E** [↗](#)).

Assessing PC inputs to the LC and mesencephalic trigeminal neurons

There are many types of molecularly distinct neurons in the brainstem ⁴⁵[↗](#), which could be differentially targeted by PCs and contribute to the highly heterogeneous amplitudes of evoked PC-IPSCs we observed. We therefore took a targeted approach to record from neurons in the MEV and LC, which are distinct neuron populations with well-characterized markers that are in close proximity to PC boutons and to areas with large PC inputs (the PB and vestibular nuclei, **Fig. 4A** [↗](#)). Previous trans-synaptic tracing efforts had also suggested potential PC inputs to these cells, but functional characterizations of these connections are lacking ^{17,19-21}.

We began by studying the extremely large unipolar neurons of the MEV. These neurons are developmentally part of the periphery, gap junction coupled, exhibit a unique chloride gradient, and directly sense and control the muscles of the jaw ⁴⁶⁻⁴⁸[↗](#). We used immunohistochemistry for parvalbumin (PV) to identify MEV neurons ⁴⁷[↗](#). We measured PC-IPSCs, filled cells with biocytin, and identified these neurons based on their shape and PV expression (**Fig. 4B** [↗](#)). Recordings were made in both coronal slices (bregma -5.45 mm and -5.65 mm) and horizontal slices, but PC-IPSCs were not evoked in any of the recorded cells (**Fig. 4C** [↗](#), n=23). Thus, despite their close proximity to PC synapses, PCs do not provide a significant source of inhibition to MEV neurons.

We also used optogenetics and immunohistochemistry to examine PC-IPSCs in the LC to directly assess whether PCs provide a major source of inhibition to LC neurons, as has been proposed ¹⁹⁻²¹[↗](#). The low density of PC synapses in the LC raised doubts about whether PCs provide a prominent source of inhibition in the LC (**Fig. 4A** [↗](#)), but it was possible that PCs might synapse onto LC dendrites that extend beyond the core of the LC. We therefore directly assessed PC-LC synaptic connectivity by measuring optically-evoked PC-IPSCs using a recording electrode containing biocytin, and used immunohistochemistry to identify TH-expressing neurons of the LC

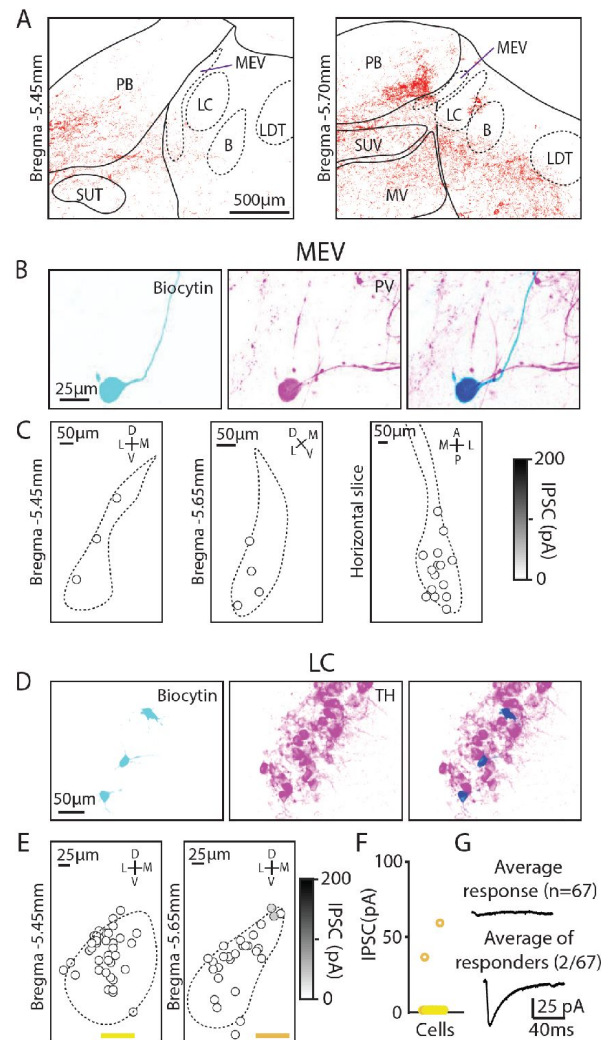


Figure 4.

Minimal PC inputs to locus coeruleus and mesencephalic trigeminal neurons

A. Example confocal image of PC presynaptic boutons labeled by synaptophysin-tdT showing cases where labeling was near locus coeruleus and the mesencephalic trigeminal nuclei.

B. (*left*) Biocytin was included in the pipette to label MEV cells during whole-cell recordings. (*middle*) MEV cells visualized with a PV antibody.

(*right*) Colocalization of biocytin and PV confirms that patched neurons were PV+ MEV neurons.

C. Locations of MEV cells in two coronal slices and one horizontal slice. No PC-IPSC responses were detected in any of the cells (n=23) at any location.

D. As in B, but for LC cells labelled with a TH antibody

E. As in C, but for two coronal levels of the LC (n=67).

F. Summary of the amplitudes of PC IPSCs recorded in LC cells.

G. Average PC-LC neuron IPSCs for all cells (n=65 *left*), and for cells where a synaptic response was detected (n=2, *right*).

(Fig. 4B). We examined the LC at two coronal levels in the posterior LC and found that no PC-IPSCs were evoked in the anterior section (0/27), while small currents were evoked in 2/40 neurons in posterior sections (bregma -5.65 mm) (Fig. 4C, D). The average PC-IPSC of all cells was 1.4 ± 8.2 pA. Thus, PCs do not provide a prominent source of inhibition to the LC.

Discussion

Widespread but nonuniform PC inputs to the brainstem

Here we find that PCs make direct synapses within numerous brainstem nuclei. The densities of these synapses vary widely in different nuclei. There is also a pronounced gradient of PC synapses across the dorsal-ventral axis of the several brainstem nuclei, indicating that PCs do not uniformly regulate activity within these nuclei. The combination of synaptophysin-TdT expression restricted to PCs and vGAT immunohistochemistry allowed us to quantify the absolute number and density of PC synapses, and the fraction of inhibitory inputs that are from PCs for each region. PC synapses comprised up to 75% of the inhibitory synapses in some vestibular regions and 55% in the caudal PB. Thus, PCs make prominent contributions to inhibition in some brainstem regions that are comparable to levels observed in cerebellar nuclei (72%)⁴⁹. In contrast, in the PRP and PCG PCs provide a low total number of inputs (PRP, PCG, vs. VN; 0.002, 0.006, >0.01 synapses/ μm^2 respectively) and a small fraction of inhibitory inputs (4, 8%, 15-50%), where they inhibit only a modest fraction of cells (13%, 33%, 54%). In all of these regions, PCs likely function similarly to their inputs to the cerebellar nuclei, where a very brief pause in firing can lead to large and rapid elevations in target cell firing^{50,51}, but PCs regulate the firing of a very different fraction of cells in brainstem nuclei and can do so in a variety of different ways^{52,53}.

PC synapses in the PCG and nearby regions

We establish that there are PC boutons within the PCG and associated nuclei, and that there are strong PC synapses in this region. This suggests that PCs play a role in regulating activity in these regions and in the behaviors they control. However, the PCG is enigmatic in structure and function, with the most commonly used mouse atlases disagreeing on the borders and identity of subnuclei^{40,54,55} (see Methods). Barrington's nucleus, a region that controls micturition^{36,37}, stands out as being well defined anatomically and functionally, and is a promising candidate for regulation by PCs. Our previous study raised the possibility that PCs in the posterior vermis might directly target cells in Barrington's nucleus¹⁷, but this study relied on AAV1 injections in the cerebellum, that in addition to anterogradely labelling PC targets, could have retrogradely labeled cells in Barrington's nucleus if they project to the cerebellar cortex. Our results highlight the challenges in studying PC synapses in the PCG and related nuclei. We observed PC synapses and synaptic connections for cells in the vicinity of Barrington's nucleus (Fig. 2 and Fig. 3). As suggestive as these findings are, it remains an open question whether direct PC synapses to Barrington's nucleus control micturition without additional experiments using molecular markers to identify Barrington's nucleus. Similarly, further studies are needed to identify other PC targets in the PCG and determine what behaviors they control.

Gradients of PC synapses in the brainstem

In general, PCs inputs to brainstem nuclei are spatially heterogeneous. The caudal-rostral gradient of PC synapses in the PB allows PCs to target specific subtypes of PB neurons¹⁷. We find that PC synapses in the PCG and nearby subnuclei also exhibit a caudal-rostral gradient. Additional studies are needed to identify PC targets in these regions and to determine the significance of their spatial heterogeneity.

There is also a pronounced dorsal-ventral gradient in many vestibular subnuclei and related nuclei. Previous studies suggested that PCs do not project to all regions of the VN, but they were unable to quantify the regional densities of PC synapses. Labeling individual PCs or those within subregions of the cerebellar cortex helped to determine their specific projections in the VN^{3,16,56-58}, but did not label all PC synapses in the VN. Immunohistochemical methods targeting proteins selectively expressed in PCs offer the potential to label all PC synapses in the VN. Calbindin is a widely used PC marker, but its presence in vestibular afferents compromises its use in the VN⁵⁹. PKC γ is also expressed by PCs, but has the advantage that it is not present in vestibular afferents⁴. The lack of PKC γ immunoreactivity established that PC synapses are absent in the ventral VN, which is qualitatively similar to what we observe. However, the possibility that PKC γ is expressed by other cell types projecting to the VN complicates the use of PKC γ labeling as a definitive marker for identifying PC synapses. Our studies establish that PC synapses are present at high densities throughout the SUV; in the dorsal LAV, MV, PRP, SPIV; in subregions of the ECU; and there are large subregions that are devoid of PC synapses. Multiple lines of evidence suggest that PC synapses in the VN play a role in cancelling self-motion related to eye movements and proprioception¹⁰. The distribution of synapses suggests that processing within VN subnuclei is spatially segregated according to the locations of PC synapses, implying that PC feedback potentially cancels signals related to self-motion specifically within these regions.

Brainstem regulation by direct PC inputs or indirectly via the deep cerebellar nuclei

PCs can also disynaptically influence the brainstem by disinhibiting cells in the deep cerebellar nuclei that in turn project to the brainstem. It is likely that some brainstem regions receive both direct PC input and indirect from deep cerebellar nuclei^{22,60,61}, and that these two different pathways serve different roles. For example, direct and indirect inputs target different regions of the PB are implicated in different behaviors^{17,62}. The deep cerebellar nuclei projections to the vestibular nuclei are not well understood, but they are likely a mix of excitatory and inhibitory inputs^{60,61,63}. This raises the possibility that the indirect pathway through the deep cerebellar nuclei has the added flexibility of providing both excitation and inhibition.

Lack of significant PC inputs to LC neurons

We find that PCs do not provide a significant direct input to the LC. This has been controversial because retrograde labelling studies using rabies viruses concluded that PCs constitute one of the largest groups of cells projecting to the LC¹⁹ (Fig 2h,i), even though PC synapses did not appear to innervate the core of the LC¹⁷ and Fig. 1, Fig. 2). We directly assessed PC to LC connection strengths using a cre line that is highly selective for PCs, and we found that in *PC/ChR2-YFP* mice, tiny optically-evoked PC-IPSCs (~50 pA) were observed in just 3% of LC neurons. The average IPSC size for all cells was less than 2 pA. This indicates that within the LC, inhibition originates from brain regions other than PCs²¹. The cerebellum may still regulate firing in the LC, but it does so via connections through the deep cerebellar nuclei and the LC^{24,30,31}.

Together, these findings indicate that reliance on anatomical tracing experiments alone is insufficient to establish the presence and importance of a synaptic connection. Studies highlighting PC inputs to the LC were conducted using rabies-tracing methods^{19,21}, which have limitations. These viruses have unknown tropism, unclear synapse specificity, and critically depend on the degree to which the “starter” cell population is restricted to the intended cell type, which can be complicated by inappropriate expression, and by the sensitivity of TVA-based rabies infection^{64,65}. We were fortunate to have a highly selective cre line to selectively target PCs that allowed us to examine PC synapses in the brainstem while avoiding complications that can occur when using viruses.

PC targets in the brainstem

The next step in clarifying the PC to brainstem outputs and the behaviors they regulate is to determine the molecular identity of target cells. This is challenging because there are many types of neurons present in the brainstem^{45,66-68}. We previously identified candidate PC targets by injecting AAV1 in the cerebellar cortex to anterogradely label PC targets, and then using RNA-seq to determine the molecular identity of labelled cells¹⁷. An important caveat to this approach is that while it very effectively labels anterograde PC targets, it can also retrogradely label cells that project to the cerebellar cortex. It is therefore crucial to determine if functional PC synapse are present, unless it is known that the region does not project to the cerebellar cortex.

Our primary focus in our previous study¹⁷ was to identify PC targets in the PB. The fact that the PB does not project to the posterior vermis of the cerebellum^{69,70}, and the observation that labelled cells were restricted to PB regions receiving PC inputs, suggest that the following cells are PC targets: *Satb2*, *Penk*, and *Tacr1* populations, which have roles in taste^{71,72}, thermoregulation⁷³, and pain^{74,75}, respectively. We also found that CGRP neurons^{70,76,77}, which are located in a region of the PB devoid of PC synapses, were not labelled by this method. This supports the hypothesis that PCs selectively inhibit specific types of brainstem neurons.

We also labelled neurons in the vestibular nucleus, Barrington's nucleus, mesencephalic trigeminal nucleus, and the locus coeruleus in that previous study. The lack of PC-evoked inputs to neurons in the mesencephalic trigeminal nucleus and locus coeruleus shown here established that these populations are not targeted by PCs. It further suggests that these neurons project to the cerebellum and were retrogradely labelled^{78,79}. For the multiple candidate PC targets in the vestibular nucleus, it is necessary to specifically test for the presence of PC synapses, as was done for floccular-targeting, *GlyT2* expressing neurons^{5,6}. In the PCG and nearby nuclei, the sparsity of PC synapses combined with the large PC synaptic inputs in a small fraction of cells suggests that PCs only inhibit specific cells in these regions. This was also reflected in our previous anatomical experiments¹⁷, where we identified *Crh*-expressing Barrington's neurons and other glutamatergic/GABAergic neurons in the PCG. These cells are important for micturition³⁶ and valence³³ respectively, and further indicate a broad range of PC output functions.

Methods

Animals

For anatomical studies, B6.Cg-Tg(PCP2-cre)3555Jdhu/J x B6;129S-Gt(ROSA)26Sor^{tm34.1}(CAG-Syp/tdTomato)Hze/J (*PC/synaptophysin-tdTomato*, Jackson Labs) was used. Calca-GFP (*CGRP-GFP*) (MMRRC, Stock number: 011187-UCD, generously provided by David Ginty) mice were crossed with the *PC/synaptophysin-tdTomato* mice to further support anatomical studies. For physiology experiments, B6.Cg-Tg(PCP2-cre)3555Jdhu/J x B6;129S-Gt(ROSA)26Sor^{tm32}(CAG-COP4*H134R/EYFP)Hze/J (*PC/ChR2-YFP*, Jackson Laboratory) mice were used. All animals were used under supervision of Harvard Medical School's Institutional Animal Care and Use Committee (IACUC) or the Pennsylvania State University College of Medicine's IACUC.

Anatomy

In order to identify inhibitory synapses PCs made in the brainstem, *PC/synaptophysin-tdTomato* mice were crossed with *CGRP-GFP* mice. Thus, synaptophysin-tdTomato was expressed selectively in all PCs and GFP is expressed in the lateral parabrachial nuclei, the LC, the hypoglossal nuclei, the vestibular nerve, the spinal tract of the trigeminal nerve, and the solitary tracts. Mice were anesthetized at P56 and perfused with PBS and 4% PFA. After removal, brains were post-fixed for

one day in 4% PFA, and sliced coronally and sagittally in 50 μm sections. Slices were stained with a vGAT (vesicular GABA transporter) antibody (Synaptic Systems, #131 004, 1:500), and visualized using an AlexaFluor 647 secondary (Thermo Fisher, #A-21450, 1:1000). Every third section of the dorsal brainstem was imaged on a confocal microscope (Leica Stellaris 5). Acquisitions were made with an XY resolution of 361.1 nm for an area of about 10 mm² total per section, and 300 nm z steps for 4 μm stacks. Stitched images were imported into ZEISS arivis Pro (arivis).

To identify PC synapses, we analyzed for synaptophysin-tdTomato fluorescence colabeled with vGAT fluorescence^{17,80}. Using a background correction feature in arivis, a background representation was generated with a 50 μm diameter Gaussian filter and subtracted from the vGAT fluorescence. vGAT was then isolated by a watershed algorithm provided with a 1 μm bouton diameter and threshold intensity. Thresholds were adjusted for each individual image based on three 50-pixel squares that had various densities of vGAT expression. Optimal threshold values were approximately one standard deviation above median. The following analysis were conducted in MATLAB: segmented vGAT boutons were analyzed for tdTomato-synaptophysin colocalization by finding a regional maximum in the tdTomato fluorescence (maximum center-to-center distance is 0.7 μm). We plotted the synaptophysin intensity values within each voxel. This histogram showed a bimodal distribution corresponding to noise and signal, and the signal threshold corresponded to the minimum between the two peaks.

We identified brainstem nuclei by aligning the Allen Common Coordinate Framework. We used several features to register confocal images of the brainstem: 1) the edges of the tissue and position of the ventricles, 2) the fiber tracts and features notable from autofluorescence and brightfield imaging, and 3) GFP labelling present in numerous brainstem regions in CGRP-GFP mice (including VIIIn, sptv, LC, XII, and their axons). CGRP-GFP labelling in the LC and VIIIn was particularly useful in anterior sections.

Puncta within each brainstem region were quantified using this aligned atlas. We matched our samples to the indicated bregma levels (± 100 μm) in **Figure 2**. To determine the variability in our alignment, we sampled 50-70 points 350 μm apart across the atlas image before and after aligning to each confocal image. The displacement distances of these reference points were calculated for each image and binned based on the confocal image location relative to bregma. Our images were displaced from the Allen Common Coordinate Framework by an average of 107 μm . The displacement distances were 90 ± 25 μm (standard deviation, n=6 hemispheres, bregma -5.45 mm), 126 ± 26 μm (n=10, bregma -5.65 mm), 102 ± 35 μm (n=8, bregma -6.05 mm), 103 ± 51 μm (n=8, bregma -6.35 mm), 108 ± 43 μm (n=8, bregma -6.65 mm), 96 ± 31 μm (n=8, bregma -6.95 mm).

It is important to note that the accuracy of our mapping is dependent on the accuracy of the reference maps that we register to. The PCG is one region in particular that varies substantially between reference atlases^{40,54,55}. We labelled the regions in/around the dorsomedial pons as the PCG, based off of the borders noted in the Allen CCF, but other atlases constrain the borders of the PCG, leaving parts of the dorsomedial pons effectively unlabelled. The precise location of many other regions within the dorsomedial pons also differ dramatically. Precise identification of these areas (i.e. Barrington's nucleus) will require a molecularly targeted approach (*in situ* hybridization, etc.).

Electrophysiology

Mice of both sexes ranging from p24 to p50 (n=30) were anesthetized with ketamine / xylazine and transcardially perfused with warm choline ACSF solution (34°C) containing in mM: 110 Choline Cl, 2.5 KCl, 1.25 NaH₂PO₄, 25 NaHCO₃, 25 glucose, 0.5 CaCl₂, 7 MgCl₂, 3.1 Na Pyruvate, 11.6 Na ascorbate, 0.002 (R,S)-CPP, 0.005 NBQX, oxygenated with 95% O₂ / 5% CO₂. Coronal slices were made (200 μm) using a Leica 1200S or a Campden Instrument 7000smz-2 vibratome in warm choline ACSF (34°C). Slices were then transferred to a holding chamber with warm ACSF (34°C)

containing in mM: 127 NaCl, 2.5 KCl, 1.25 NaH₂PO₄, 25 NaHCO₃, 25 glucose, 1.5 CaCl₂, 1 MgCl₂ and were recovered at 34°C for 30 minutes before being moved to room temperature until recordings began.

Voltage-clamp recordings were made across the brainstem. Borosilicate glass electrodes (2–4 MΩ) were filled with a high chloride internal containing in mM: 35 CsF, 100 CsCl, 10 EGTA, 10 HEPES, 4 QX-314, pH 7.3 with CsOH. This internal solution was used to increase sensitivity to inhibitory inputs and improve clamp. The osmolarity of internal solution was adjusted to 290–300 mOsm. Whole-cell capacitance and series resistance was left uncompensated for all experiments. Neurons were held at -70 mV. All experiments were performed at room temperature in the presence of 5 μM NBQX to block AMPARs and 2.5 μM (R,S)-CPP to block NMDARs, with a flow rate of 2 ml/min. ChR2 expressing PC axons were stimulated with 473 nm light from a light-emitting diode (Thorlabs or Sutter) in the entire field of view (1 ms, ~80 mW/mm²) for at least 10 trials. Data were collected using Igor (Wavemetrics) running mafPC (courtesy of M. A. Xu-Friedman) or SutterPatch (Sutter). Images with a 4X objective were used to determine the positions of recorded cells. Slices were fixed and stained with DAPI to confirm the anterior-posterior position of the coronal slice, and matched to the closest indicated bregma level indicated in [Figure 3](#).

In experiments targeting the LC or mesencephalic trigeminal nucleus, biocytin (B1592, Thermofisher) was included in the intracellular solution. Cells were then stained with an anti-tyrosine hydroxylase antibody to identify the LC (1:1000, AB152, Sigma) or anti-parvalbumin antibody to identify the mesencephalic trigeminal (1:1000 P3088, Sigma), followed by an Alexa 647 secondary antibody (1:1000 A32733 or ab150079, Thermofisher or Abcam respectively) and streptavidin-Alexa 594 conjugate (1:1000, 1402-3900, Thermofisher). Sections were subsequently imaged using a confocal microscope (Leica Stellaris 5) to determine the identity of the recorded cell.

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	Nuclei		
B	Barrington's Nucleus	PRP	Prepositus Nucleus
CU	Cuneate Nucleus	SG	Supragenual Nucleus
DCO	Dorsal Cochlear Nucleus	SLD	Sublaterodorsal Nucleus
DTN	Dorsal Tegmental Nucleus	SPIV	Spinal Vestibular Nucleus
ECU	External Cuneate Nucleus	SPVI	Spinal Nucleus of the Trigeminal
GRN	Gigantocellular Reticular Nucleus	SUT	Supratrigeminal Nucleus
IP	Interposed Nucleus	SUV	Superior Vestibular Nucleus
LAV	Lateral Vestibular Nucleus	VN	Vestibular Nucleus
LC	Locus Coeruleus	X	Nucleus X
LDT	Laterodorsal Tegmental Nucleus	XII	Hypoglossal Nucleus
MEV	Mesencephalic Trigeminal Nucleus	Y	Nucleus Y
MV	Medial Vestibular Nucleus		
NI	Nucleus Incertus		Ventricles
NTS	Nucleus of the Solitary Tract	V4	Fourth Ventricles
NTSI	Nucleus of the Solitary Tract, lateral	V4r	Fourth Ventricle (Lateral Recess)
NTSm	Nucleus of the Solitary Tract, medial		Fiber Tracts
PARN	Parvicellular Reticular Nucleus	arb	Arbor vitae
PB	Parabrachial Nuclei	icp	Inferior cerebellar peduncle
PCG	Pontine Central Gray	scp	Superior cerebellar peduncle
PGRN	Paragigantocellular Reticular Nucleus	sptv	Spinal tract of the trigeminal nerve
PGRNd	Paragigantocellular Reticular Nucleus, dorsal	VIIIn	Facial nerve
PRNr	Pontine Reticular Nucleus	vVIIIIn	Vestibular nerve

Table 1.

Abbreviations of Brainstem Anatomical Regions

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Author information

Christopher H Chen*

Department of Neurobiology, Harvard Medical School, Boston, United States, Department of Neural and Behavioral Sciences, The Pennsylvania State University, Hershey, United States

For correspondence: chc5230@psu.edu

*These authors contributed equally

Zhiyi Yao*

Department of Neurobiology, Harvard Medical School, Boston, United States, Department of Neural and Behavioral Sciences, The Pennsylvania State University, Hershey, United States

*These authors contributed equally

Shuting Wu

Department of Neurobiology, Harvard Medical School, Boston, United States
ORCID iD: [0000-0003-2264-7393](https://orcid.org/0000-0003-2264-7393)

Wade G Regehr

Department of Neurobiology, Harvard Medical School, Boston, United States

ORCID iD: [0000-0002-3485-8094](https://orcid.org/0000-0002-3485-8094)

For correspondence: wade_regehr@hms.harvard.edu

Editors

Reviewing Editor

Aya Ito-Ishida

RIKEN Center for Brain Science, Saitama, Japan

Senior Editor

Lu Chen

Stanford University, Stanford, United States of America

Reviewer #1 (Public review):

Summary:

This paper is an incremental follow-up to the authors' recent paper which showed that Purkinje cells make inhibitory synapses onto brainstem neurons in the parabrachial nucleus which project directly to the forebrain. In that precedent paper, the authors used a mouse line which expresses the presynaptic marker synaptophysin in Purkinje cells to identify Purkinje cell terminals in the brainstem and they observed labeled puncta not only in the vestibular and parabrachial nuclei, as expected, but also in neighboring dorsal brainstem nuclei, prominently the central pontine grey. The present study, motivated by the lack of thorough characterization of PC projections to brainstem, uses the same mouse line to anatomically map the density and a PC-specific channelrhodopsin mouse line to electrophysiologically assess the strength of Purkinje cell synapses in dorsal brainstem nuclei. The main findings are (1) the density of Purkinje cell synapses is highest in vestibular and parabrachial nuclei and correlates with the magnitude of evoked inhibitory synaptic currents, and (2) Purkinje cells also synapse in the central pontine grey nucleus but not in the locus coeruleus or mesencephalic nucleus.

Strengths:

The complementary use of anatomical and electrophysiological methods to survey the distribution and efficacy of Purkinje cell synapses on brainstem neurons in mouse lines that express markers and light-sensitive opsins specifically in Purkinje cells is the major strength of this study. By systematically mapping presynaptic terminals and light-evoked inhibitory postsynaptic currents in dorsal brainstem, the authors provide convincing evidence that Purkinje cells do synapse directly onto pontine central grey and nearby neurons but do not synapse onto trigeminal motor or locus coeruleus neurons. Their results also confirm previously documented heterogeneity of Purkinje cell inputs to vestibular nucleus and parabrachial neurons.

Weaknesses:

Although the study provides strong evidence that Purkinje cells do not make extensive synapses onto LC neurons, which is a helpful caveat given previous reports to the contrary, it falls short of providing the comprehensive characterization of Purkinje cell brainstem synapses which seemed to be the primary motivation of the study. The main information provided is a regional assessment of PC density and efficacy, which seems of limited utility

given that we are not informed about the different sources of PC inputs, variations in the sizes of PC terminals, the subcellular location of synaptic terminals, or the anatomical and physiological heterogeneity of postsynaptic cell types. The title of this paper would be more accurate if "characterization" were replaced by "survey".

Several of the study's conclusions are quite general and have already been made for vestibular nuclei, including the suggestions in Abstract, Results, and Discussion that PCs selectively influence brainstem subregions and that PCs target cell types with specific behavioral roles.

<https://doi.org/10.7554/eLife.101825.2.sa3>

Reviewer #2 (Public review):

Summary:

While it is often assumed that the cerebellar cortex connects, via its sole output neuron, Purkinje cell, exclusively to the cerebellar nuclei, axonal projections of the Purkinje cells to dorsal brainstem regions have been well documented. This paper provides comprehensive mapping and quantification of such extracerebellar projections of the Purkinje cells, most of which are confirmed with electrophysiology in slice preparation. A notable methodological strength of this work is the use of highly Purkinje cell-specific transgenic strategies, enabling selective and unbiased visualization of Purkinje terminals in the brainstem. By utilizing these selective mouse lines, the study offers compelling evidence challenging the general assumption that Purkinje cell targets are limited to the cerebellar nuclei. While the individual connections presented are not entirely novel, this paper provides a thorough and unambiguous demonstration of their collective significance. Regarding another major claim of this paper, "characterization of direct Purkinje cell outputs (Title)", however, the depth of electrophysiological analysis is limited to presence/absence of physiological Purkinje input to postsynaptic brainstem neurons whose known cell types are mostly blinded. Overall, conceptual advance is largely limited to confirmatory or incremental, although it would be useful for the field to have the comprehensive landscape presented.

Strengths:

Unsupervised comprehensive mapping and quantification of the Purkinje terminals in the dorsal brainstem are enabled, for the first time, by using the current state-of-the-art mouse lines, BAC-Pcp2-Cre and synaptophysin-tdTomato reporter (Ai34).

Combinatorial quantification with vGAT puncta and synaptophysin-tdTomato labeled Purkinje terminals clarifies the anatomical significance of the Purkinje terminals as an inhibitory source in each dorsal brainstem region.

Electrophysiological confirmation of the presence of physiological Purkinje synaptic input to 7 out of 9 dorsal brainstem regions identified.

Pan-Purkinje ChR2 reporter provides solid electrophysiological evidence to help understand the possible influence of the Purkinje cells onto LC.

Weaknesses:

The present paper is largely confirmatory to what is presented in a previous paper published by the author's group (Chen et al., 2023, Nat Neurosci). In this preceding paper, the author's group used AAV1-mediated anterograde transsynaptic strategy to identify postsynaptic neurons of the Purkinje cells. The experiments performed in the present paper is, by nature, complementary to the AAV1 tracing which can also infect retrogradely and thus is not able to

demonstrate the direction of synaptic connections between reciprocally connected regions. Anatomical findings are all consistent with the preceding paper.

While the authors appear to assume uniform cell type and postsynaptic response in each of the dorsal brainstem nuclei (as noted in the Discussion, "PCs likely function similarly to their inputs to the cerebellar nuclei, where a very brief pause in firing can lead to large and rapid elevations in target cell firing"), we know that the responses to the Purkinje cell input are cell type dependent, which vary in neurotransmitter, output targets, somata size, and distribution, in the cerebellar and vestibular nuclei (Shin et al., 2011, J Neurosci; Najac and Raman, 2015, J Neurosci; Özcan et al., 2020, J Neurosci). Also, whether 23 % (for PCG), for example, is "a small fraction" would be subjective: it might represent a numerically small but functionally important cell type population. From a functional perspective, the cell type-blind physiological characterization in this manuscript remains superficial compared to existing cell type-specific analyses, although the authors commented on these issues in the manuscript.

<https://doi.org/10.7554/eLife.101825.2.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript by Chen and colleagues explores the connections from cerebellar purkinje cells to various brainstem nuclei. They combine two methods - presynaptic puncta labeling as putative presynaptic markers, and optogenetics, to test the anatomical projections and functional connectivity from purkinje cells onto a variety of brainstem nuclei. Overall, their study provides an atlas of sorts of purkinje cell connectivity to the brainstem, which includes a critical analysis of some of their own data from another publication. Overall, the value of this work is to both provide neural substrates by which purkinje cells may influence the brainstem and subsequent brain regions independent of the deep cerebellar nuclei, and also, to provide a critical analysis of viral-based methods to explore neuronal connectivity.

Strengths:

The strengths lie in the simplicity of the study, the number of cells patched, and the relationship between the presence of putative presynaptic puncta and electrophysiological results. This type of study is important and should provide a foundation for future work exploring cerebellar inputs and outputs. Overall, I think that the critique of viral-based methods to define connectivity, and a more holistic assessment of what connectivity is and how it should be defined is timely and warranted, as I think this is under-appreciated by many groups and overall, there is a good deal of research being published that do not properly consider the issues that this manuscript raises about what viral-based connectivity maps do and do not tell us.

Weaknesses:

While I overall liked the manuscript, I do have a few concerns which relate to interpretation of results, and discussion of technological limitations. The main concerns I have relate to the techniques that the authors use, and an insufficient discussion of their limitations. The authors use a Cre-dependent mouse line that expresses a synaptophysin-tdtomato marker, which the authors confidently state is a marker of synapses. This is misleading. Synaptophysin is a vesicle marker, and as such, labels axons, where vesicles are present in transit, and likely cell bodies where the protein is being produced. As such, the presence of tdtomato should not be interpreted definitively as the presence of a synapse. The use of vGAT as a marker, while this helps to constrain the selection of putative pre-synaptic sites, is also a

vesicle marker and will likely suffer the same limitations (though in this case the expression is endogenous and not driven by the ROSA locus). A more conservative interpretation of the data would be that the authors are assessing putative pre-synaptic sites with their analysis. This interpretation is wholly consistent with their findings showing the presence of tdtomato in some regions but only sparse connectivity - this would be expected in the event that axons are passing through. If the authors wish to strongly assert that they are specifically assessing synapses, a marker better restricted to synapses and not vesicles may be more appropriate.

Similarly, while optogenetics/slice electrophysiology remains the state of the art for assessing connectivity between cell populations, it is not without limitations. For example, connections that are not contained within the thickness of the slice (here, 200 μ m, which is not particularly thick for slice ephys preps) will not be detected. As such, the absence of connections are harder to interpret than the presence of connections. Slices were only made in the coronal plane, which means if that if there is a particular topology to certain connections that is orthogonal to that plane, those connections may be under-represented. As such, all connectivity analyses likely are under-representations of the actual connectivity that exists in the intact brain. Therefore, perhaps the authors should consider revising their assessments of connections, or lack thereof, of purkinje cells to e.g., LC cells. While their data do make a compelling case that the connections between purkinje cells and LC cells are not particularly strong or numerous, especially compared to other nearby brainstem nuclei, their analyses do indicate that at least some such connections do exist. Thus, rather than saying that the viral methods such as rabies virus are not accurate reflections of connectivity - perhaps a more circumspect argument would be that the quantitative connectivity maps reported by other groups using rabies virus do not always reflect connectivity defined by other means e.g., functional connections with optogenetics. In some cases the authors do suggest this (e.g., "Together, these findings indicate that reliance on anatomical tracing experiments alone is insufficient to establish the presence and importance of a synaptic connection"), but in other cases they are more dismissive of viral tracing results (e.g., "it further suggests that these neurons project to the cerebellum and were not retrogradely labeled"). Furthermore, some statements are a bit misleading e.g., mentioning that rabies methods are critically dependent on starter cell identity immediately following the citation of studies mapping inputs onto LC cells. While in general this claim has merit, the studies cited (19-21) use Dbh-Cre to define LC-NE cells which does have good fidelity to the cells of interest in the LC. Therefore, rewording this section in order to raise these issues generally without proximity to the citations in the previous sentence may maintain the authors' intention without suggesting that perhaps the rabies studies from LC-NE cells that identified inputs from purkinje cells were inaccurate due to poor fidelity of the Cre line. Overall, this manuscript would certainly not be the first report indicating that rabies virus does not provide a quantitative map of input connections. In my opinion this is still under-appreciated by the broad community and should be explicitly discussed. Thus, an acknowledgement of previous literature on this topic and how their work contributes to that argument is warranted.

Comments on revisions:

The responses the authors offer in theory are good, but they still use terms such as synapses and putative presynaptic boutons relatively interchangeably - if the authors make the correction to the more conservative terminology, which I think better reflects the data, this should be more consistent throughout the manuscript.

<https://doi.org/10.7554/eLife.101825.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This paper is an incremental follow-up to the authors' recent paper which showed that Purkinje cells make inhibitory synapses onto brainstem neurons in the parabrachial nucleus which project directly to the forebrain. In that precedent paper, the authors used a mouse line that expresses the presynaptic marker synaptophysin in Purkinje cells to identify Purkinje cell terminals in the brainstem and they observed labeled puncta not only in the vestibular and parabrachial nuclei, as expected, but also in neighboring dorsal brainstem nuclei, prominently the central pontine grey. The present study, motivated by the lack of thorough characterization of PC projections to the brainstem, uses the same mouse line to anatomically map the density and a PC-specific channelrhodopsin mouse line to electrophysiologically assess the strength of Purkinje cell synapses in dorsal brainstem nuclei. The main findings are (1) the density of Purkinje cell synapses is highest in vestibular and parabrachial nuclei and correlates with the magnitude of evoked inhibitory synaptic currents, and (2) Purkinje cells also synapse in the central pontine grey nucleus but not in the locus coeruleus or mesencephalic nucleus.

Strengths:

The complementary use of anatomical and electrophysiological methods to survey the distribution and efficacy of Purkinje cell synapses on brainstem neurons in mouse lines that express markers and light-sensitive opsins specifically in Purkinje cells is the major strength of this study. By systematically mapping presynaptic terminals and light-evoked inhibitory postsynaptic currents in the dorsal brainstem, the authors provide convincing evidence that Purkinje cells do synapse directly onto pontine central grey and nearby neurons but do not synapse onto trigeminal motor or locus coeruleus neurons. Their results also confirm previously documented heterogeneity of Purkinje cell inputs to the vestibular nucleus and parabrachial neurons.

Weaknesses:

Although the study provides strong evidence that Purkinje cells do not make extensive synapses onto LC neurons, which is a helpful caveat given previous reports to the contrary, it falls short of providing the comprehensive characterization of Purkinje cell brainstem synapses which seemed to be the primary motivation of the study. The main information provided is a regional assessment of PC density and efficacy, which seems of limited utility given that we are not informed about the different sources of PC inputs, variations in the sizes of PC terminals, the subcellular location of synaptic terminals, or the anatomical and physiological heterogeneity of postsynaptic cell types. The title of this paper would be more accurate if "characterization" were replaced by "survey".

Several of the study's conclusions are quite general and have already been made for vestibular nuclei, including the suggestions in the Abstract, Results, and Discussion that PCs selectively influence brainstem subregions and that PCs target cell types with specific behavioral roles.

We agree that we did not provide an in-depth characterization of PC synapses onto all identified types of brainstem neurons. With so many types of neurons in the brainstem, this would be a monumental task. Despite this limitation we prefer to keep our original title, since our study makes the following advances:

- We provide a comprehensive map of all PC synaptic boutons across the brainstem, and corresponding maps of PC synaptic input sizes. The input sizes vary widely, but are often multiple nanoamps, indicating that the cerebellum is an important regulator of activity in these regions. These maps will be indispensable for future investigations of cerebellar outputs.
- We find that PC projections and the synapses they make are spatially restricted within most target nuclei such as the vestibular and parabrachial nuclei. This suggests that the influence of the cerebellum is spatially segregated within these nuclei, and likely allows the cerebellum to regulate specific behaviors. While some aspects of these gradients have been described previously, our study is comprehensive, and has a higher degree of specificity than can be achieved with immunohistochemistry.
- We discover that PCs form functional synapses in the pontine central grey and nearby nuclei. Much of this region's function is unknown, but certain subregions are important for micturition and valence. PCs make large synapses onto a small fraction of cells in this region, which suggests that PCs may target specific cell types to control novel nonmotor behaviors.
- We provide clarification regarding PC projections to the locus coeruleus. Multiple high-profile, highly influential studies using rabies tracing (Schwarz et al., *Nature* 2015; Breton-Provencher and Sur, *Nature Neuroscience* 2019; and others) described a prominent PC input to the locus coeruleus. We showed that this projection is essentially nonexistent, both anatomically and functionally. We previously addressed this issue, but the PC-specific optogenetic approach we used here provides the most compelling evidence against a prominent PC-LC connection. This is an important finding for the cerebellum and a cautionary tale for conclusions based solely on viral tracing methods. We will expand on this issue in response to the comments of reviewer #3.

Reviewer #2 (Public review):

Summary:

While it is often assumed that the cerebellar cortex connects, via its sole output neuron, the Purkinje cell, exclusively to the cerebellar nuclei, axonal projections of the Purkinje cells to dorsal brainstem regions have been well documented. This paper provides comprehensive mapping and quantification of such extracerebellar projections of the Purkinje cells, most of which are confirmed with electrophysiology in slice preparation. A notable methodological strength of this work is the use of highly Purkinje cell-specific transgenic strategies, enabling selective and unbiased visualization of Purkinje terminals in the brainstem. By utilizing these selective mouse lines, the study offers compelling evidence challenging the general assumption that Purkinje cell targets are limited to the cerebellar nuclei. While the individual connections presented are not entirely novel, this paper provides a thorough and unambiguous demonstration of their collective significance. Regarding another major claim of this paper, "characterization of direct Purkinje cell outputs (Title)", however, the depth of electrophysiological analysis is limited to the presence/absence of physiological Purkinje input to postsynaptic brainstem neurons whose known cell types are mostly blinded. Overall, conceptual advance is largely limited to confirmatory or incremental, although it would be useful for the field to have the comprehensive landscape presented.

Strengths:

(1) Unsupervised comprehensive mapping and quantification of the Purkinje terminals in the dorsal brainstem are enabled, for the first time, by using the current state-of-the-art mouse lines, BAC-Pcp2-Cre and synaptophysin-tdTomato reporter (Ai34).

(2) Combinatorial quantification with vGAT puncta and synaptophysin-tdTomato labeled Purkinje terminals clarifies the anatomical significance of the Purkinje terminals as an inhibitory source in each dorsal brainstem region.

(3) Electrophysiological confirmation of the presence of physiological Purkinje synaptic input to 7 out of 9 dorsal brainstem regions identified.

(4) Pan-Purkinje ChR2 reporter provides solid electrophysiological evidence to help understand the possible influence of the Purkinje cells onto LC.

Weaknesses:

(1) The present paper is largely confirmatory of what is presented in a previous paper published by the author's group (Chen et al., 2023, Nat Neurosci). In this preceding paper, the author's group used AAV1-mediated anterograde transsynaptic strategy to identify postsynaptic neurons of the Purkinje cells. The experiments performed in the present paper are, by nature, complementary to the AAV1 tracing which can also infect retrogradely and thus is not able to demonstrate the direction of synaptic connections between reciprocally connected regions. Anatomical findings are all consistent with the preceding paper. The likely absence of robust physiological connections from the Purkinje to LC has also been evidenced in the preceding paper by examining c-Fos response to Purkinje terminal photoinhibition at the PBN/LC region.

We agree that we previously dealt with the issue of PC-LC synapses (Chen et al., 2023, Nat Neurosci), but our conclusions differed from several high-profile publications (Schwarz et al., Nature 2015; Breton-Provencher and Sur, Nature Neuroscience 2019), and still met considerable resistance. We felt that the optogenetic approach provided the most definitive means of evaluating the presence and strength of PC-LC synapse that will hopefully settle this issue. These experiments also set a standard for future studies assessing the presence of PC synapses onto other target neurons in the brainstem.

(2) Although the authors appear to assume uniform cell type and postsynaptic response in each of the dorsal brainstem nuclei (as noted in the Discussion, "PCs likely function similarly to their inputs to the cerebellar nuclei, where a very brief pause in firing can lead to large and rapid elevations in target cell firing"), we know that the responses to the Purkinje cell input are cell type dependent, which vary in neurotransmitter, output targets, somata size, and distribution, in the cerebellar and vestibular nuclei (Shin et al., 2011, J Neurosci; Najac and Raman, 2015, J Neurosci; Özcan et al., 2020, J Neurosci). This consideration impacts the interpretation of two key findings: (a) "Large ... PC-IPSCs are preferentially observed in subregions with the highest densities of PC synapses (Abstract)". For example, we know that the terminal sparse regions reported in the present paper do contain Floccular Targeted Neurons that are sparse yet have dense somatic terminals with profound postinhibitory rebound (Shin et al.). Despite their sparsity, these postsynaptic neurons play a distinct and critical role in proper vestibuloocular reflex. Therefore, associating broad synaptic density with "PC preferential" targets, as written in the Abstract, may not fully capture the behavioral significance of Purkinje extracerebellar projections. (b) "We conclude ... only a small fraction of cell. This suggests that PCs target cell types with specific behavioral roles (Abstract, the last sentence)". Prior research has already established that "PCs target cell types with specific behavioral roles in brainstem regions". Also, whether 23 % (for PCG), for example, is "a small fraction" would be subjective: it might represent a numerically small but functionally important cell type population. The physiological characterization provided in the present cell type-blind analysis could, from a functional perspective, even be decremental when compared to existing cell typespecific analyses of the Purkinje cell inputs in the literature.

We now cite the papers suggested by the reviewer (Shin et al., 2011, J Neurosci; Najac and Raman, 2015, J Neurosci; Özcan et al., 2020, J Neurosci) and add to the discussion.

(3) The quantification analyses used to draw conclusions about

(a) the significance of PC terminals among all GABAergic terminals and the fractions of electrophysiologically responsive postsynaptic brainstem neurons may have potential sampling considerations:.

(a.i) this study appears to have selected subregions from each brainstem nucleus for quantification (Figure 2). However, the criteria for selecting these subregions are not explicitly detailed, which could affect the interpretation of the results.

Additional explanation has been added to results in the section, “Quantification of PC synapses in the brainstem.”

(a.ii) the mapping of recorded cells (Figure 3) seems to show a higher concentration in terminal-rich regions of the vestibular nuclei.

In Figure 3, we strived to record in an unbiased manner. However, there may have been a slight bias to recordings in areas of lower myelination where patching is easier. We now clarify this issue in the text.

Reviewer #3 (Public review):

Summary:

The manuscript by Chen and colleagues explores the connections from cerebellar Purkinje cells to various brainstem nuclei. They combine two methods - presynaptic puncta labeling as putative presynaptic markers, and optogenetics, to test the anatomical projections and functional connectivity from Purkinje cells onto a variety of brainstem nuclei. Overall, their study provides an atlas of sorts of Purkinje cell connectivity to the brainstem, which includes a critical analysis of some of their own data from another publication. Overall, the value of this work is to both provide neural substrates by which Purkinje cells may influence the brainstem and subsequent brain regions independent of the deep cerebellar nuclei and also, to provide a critical analysis of viral-based methods to explore neuronal connectivity.

Strengths:

The strengths lie in the simplicity of the study, the number of cells patched, and the relationship between the presence of putative presynaptic puncta and electrophysiological results. This type of study is important and should provide a foundation for future work exploring cerebellar inputs and outputs. Overall, I think that the critique of viral-based methods to define connectivity, and a more holistic assessment of what connectivity is and how it should be defined is timely and warranted, as I think this is under-appreciated by many groups and overall, there is a good deal of research being published that do not properly consider the issues that this manuscript raises about what viral-based connectivity maps do and do not tell us.

We thank the reviewer for highlighting this important aspect of this work, and for agreeing with our thesis concerning viral-based connectivity maps.

Weaknesses:

While I overall liked the manuscript, I do have a few concerns that relate to interpretation of results, and discussion of technological limitations. The main concerns I have relate to the techniques that the authors use, and an insufficient discussion of their limitations. The authors use a Cre-dependent mouse line that expresses a synaptophysin-tomato marker, which the authors confidently state is a marker of synapses. This is misleading. Synaptophysin is a vesicle marker, and as such, labels axons, where vesicles are present in transit, and likely cell bodies where the protein is being produced. As such, the presence of tdtomato should not be interpreted definitively as the presence of a synapse. The use of vGAT as a marker, while this helps to constrain the selection of putative pre-synaptic sites, is also a vesicle marker and will likely suffer the same limitations (though in this case, the expression is endogenous and not driven by the ROSA locus). A more conservative interpretation of the data would be that the authors are assessing putative pre-synaptic sites with their analysis. This interpretation is wholly consistent with their findings showing the presence of tdtomato in some regions but only sparse connectivity - this would be expected in the event that axons are passing through. If the authors wish to strongly assert that they are specifically assessing synapses, a marker better restricted to synapses and not vesicles may be more appropriate.

We agree that synaptophysin-tdTomato is an imperfect marker, although it is vastly superior to cytosolic tdTomato. We found that viral expression of synaptophysin-GFP gives much more punctate labelling, but an appropriate synaptophysin-GFP line is not available. We carefully point out this issue, and threshold the images to avoid faint labeling associated with fibers of passage. The intersection of VGAT labelling and of the synaptophysin-tdTomato labelling provides us with superior identification of PC boutons. We will add additional clarification to point out that these are putative presynaptic boutons, but that alone this does not establish the existence or the strength of functional synapses.

Similarly, while optogenetics/slice electrophysiology remains the state of the art for assessing connectivity between cell populations, it is not without limitations. For example, connections that are not contained within the thickness of the slice (here, 200 μ m, which is not particularly thick for slice ephys preps) will not be detected. As such, the absence of connections is harder to interpret than the presence of connections. Slices were only made in the coronal plane, which means that if there is a particular topology to certain connections that is orthogonal to that plane, those connections may be under-represented. As such, all connectivity analyses likely are under-representations of the actual connectivity that exists in the intact brain. Therefore, perhaps the authors should consider revising their assessments of connections, or lack thereof, of Purkinje cells to e.g., LC cells. While their data do make a compelling case that the connections between Purkinje cells and LC cells are not particularly strong or numerous, especially compared to other nearby brainstem nuclei, their analyses do indicate that at least some such connections do exist. Thus, rather than saying that the viral methods such as rabies virus are not accurate reflections of connectivity - perhaps a more circumspect argument would be that the quantitative connectivity maps reported by other groups using rabies virus do not always reflect connectivity defined by other means e.g., functional connections with optogenetics. In some cases, the authors do suggest this (e.g. "Together, these findings indicate that reliance on anatomical tracing experiments alone is insufficient to establish the presence and importance of a synaptic connection"), but in other cases, they are more dismissive of viral tracing results (e.g. "it further suggests that these neurons project to the cerebellum and were not retrogradely labeled"). Furthermore, some statements are a bit misleading e.g., mentioning that rabies methods are critically dependent on starter cell identity immediately following the citation of studies mapping inputs onto LC cells. While in general, this claim has merit, the studies cited (19-21) use Dbh-Cre to define LC-NE cells which does have good fidelity to the cells

of interest in the LC. Therefore, rewording this section in order to raise these issues generally without proximity to the citations in the previous sentence may maintain the authors' intention without suggesting that perhaps the rabies studies from LC-NE cells that identified inputs from Purkinje cells were inaccurate due to poor fidelity of the Cre line. Overall, this manuscript would certainly not be the first report indicating that the rabies virus does not provide a quantitative map of input connections. In my opinion, this is still under-appreciated by the broad community and should be explicitly discussed. Thus, an acknowledgment of previous literature on this topic and how their work contributes to that argument is warranted.

We have a different take on connectivity and the use of optogenetics. Based on our years of experience studying synapses in brain slice, axons survive very well even when they are cut. It is not necessary to preserve intact axons that extend for long distances. It is also true that activation of these axons, with either extracellular electrical stimulation or with optogenetics, is sufficient to evoke synaptic inputs. Robust synaptic responses are evoked with optogenetic activation regardless of the slice orientation. We thank the reviewer for raising this issue, and we have added a couple of sentences to clarify this point under the section “Characterization of functional properties of PC synapses in the brainstem.”

The discussion on starter cell specificity was not referring to the specificity of cre in transgenic animals, but the TVA/G helper proteins that are introduced by AAV and used in conjunction with the rabies virus. The issues related to this have recently been discussed in Elife (Beier, 2022) in addition to citations 58 and 59 in the manuscript. We have more explicitly highlighted this issue in the revised manuscript in the section “Lack of significant PC inputs to LC neurons.”

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Methods need detail to be replicable, particularly in how PC synapses were identified and automatically counted. It is not clear what was the variation within subregions across mice. How were neurons selected or rejected for recordings and analyses? Was each subregion sampled at equal spacing? Methods for anatomy should mention sagittal sections.

Wording in Methods section, “Anatomy” was changed to better reflect how PC synapses were identified as colabeled segments of vGAT and tdTomato labeling.

Each datapoint in Figure 2D-F was quantification of a region for each section and each mouse. The color of the data point indicates the anterior posterior location of the section. The violin plot quantifies the median and quartile value for all points across sections and mice. The variability captured by the violin point reflects variability across the anterior-posterior axis.

Neurons were mostly randomly selected in each slice, and rejected based on unstable holding current or access resistance. Cell locations were recorded and updated with each experiment so that we minimized oversampling easier to patch regions.

Sagittal sections were added in methods.

(2) Figure 2D-F what is the black line and grey region?

Additional text was added in the caption for Figure 2D-F

(3) MEV is confusing given LAV stands for lateral vestibular - perhaps call it ME5?

We will remain consistent with the abbreviations in the Allen Brain Reference Atlas.

Reviewer #2 (Recommendations for the authors):

(1) What are the criteria for distinguishing large, small, and non-responders?

Large are in the nA range, small are in the hundreds of pA, and non-responders are effectively zero. Manual curation of these responses indicated that a current amplitude threshold of 45 pA clearly separated non-responders from responders. To be clear, the average response (as stated in text and displayed in Figure 3D) includes all cells.

(2) p1. "Unexpectedly": it would not be unexpected, rather, expected, because it was reported in Chen et al., 2023, Nat Neurosci.

The PCG was hinted at, but an actual functional, anatomical connection was not reported in our previous manuscript.

(3) p1. "We combined electrophysiological recordings with immunohistochemistry to assess the molecular identities of these PC targets": please clarify "these" here. It could be read that it refers to "pontine central gray and nearby subnuclei" but it doesn't make sense. Immuno has only been performed for MeV and LC.

Corrected

(4) p1. "but only inhibit a small fraction of cells in many nuclei": as far as I read Fig.3, it seems that ~50% for PBN/VN and ~25% for PCG: would this be "a small fraction"?

The small fraction of cells was in reference to subnuclei within the PCG, but we agree this statement is too broad to be useful and have eliminated it.

(5) p2. "conventional tracer": viral tracer is becoming a standard, so dye tracer could be better here.

Corrected

(6) p3. "rostral/cauda": typo.

Corrected.

(7) p3. Quantification of PC synapses in the brainstem: it would be helpful to introduce why synapto-tdT alone is not sufficient, and the purpose of adding vGAT immunostaining.

We have added more on vGAT labeling putative presynaptic sites and quantifying only synaptic labeling instead of axonal tdTomato in the Results, "Quantification of PC synapses in the brainstem." In addition, vGAT staining allows us to examine the PC contribution to total inhibition in each region.

(8) p7. "PB and are": typo.

Corrected. And all instances of PBN were changed to PB

(9) p7. "they are likely a mix of excitatory and inhibitory inputs 54,55": Bagnall et al., 2009, J Neurosci, would be critically relevant here.

Added, thank you

| (10) Figures 2-3: Yellow/Blue color scheme is hard to distinguish, and having two colors could be read as implying two distinct regions.

We are unsure what the reviewer is referring to exactly here, but the colors refer to the sections in 2C (see the color bar on the bottom right of each atlas schematic). The points represent an individual section that was quantified, and thus do represent distinct samples from distinct regions.

| (11) Figure 2D-F: what is indicated by each point?

Each data point is the number of PC bouton (D), density of bouton (E), or percentage of synaptophysin/vGAT (F) quantified for each region per section. Each color represents a coronally distinct section of a region. Additional text was added into the captions to clarify this and point 10.

| (12) Figure 3E, right: what is the correlation coefficient?

The correlation coefficient was found to be 0.74

| **Reviewer #3 (Recommendations for the authors):**

| *Some minor grammatical errors and typos need to be cleaned up (e.g. "To quantifying the densities...", "The medial-ventral region of the PBN...have extensive...").*

These errors have been corrected

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